

A Tool for Thermal Image Annotation and Automatic Temperature Extraction around Orthopedic Pin Sites

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Abstract—Existing annotation tools are mainly designed for visible images to support supervised learning problems for machine learning. A few tools exist for extracting temperature information from thermal images. However, they are time and manpower consuming, require different stages of data management, and are not automated. This paper focuses on addressing the limitation of existing tools in handling big thermal datasets for annotation, temperature distribution extraction in the Region of Interest (ROI) of Orthopedic surgical wounds and provides flexibility for a researcher to integrate thermal image analysis into wound care machine learning models. We present an easy to use research tool for one click annotation of Orthopedic pin sites for extraction of thermal information, which is a preliminary step of research to estimate the reliability of thermography for home based surveillance of post-operative infection. The proposed tool maps annotations from visible registered image onto thermal and radiometric images. Mapping these annotations from visible registered images avoids manual bias in annotating thermal images. Integrating the functionality of an annotation tool by processing thermal images to acquire single-click manual annotations and extracting temperature distributions in the ROI with those acquired annotations is the novelty of the proposed work and is also crucial for research on deep learning-based investigation on surgical wound infections.

Index Terms—Thermal Image Annotation Tool, Temperature Exaction, Orthopedic Pin Site

I. INTRODUCTION

A pin site is a surgical site on the skin surface where a thin wire or a thicker half-pin is penetrating the skin, travelling through the deeper soft tissue layers, connecting the external frame construct to the internal bone for fixation of a fracture. At a pin site, inflammation is ongoing due to the natural healing process, but since the pin is blocking and continuously

disturbing the healing process keeping the wound open, there is a higher infection rate compared to other surgical wounds [3]. An increased rate of metabolic activity at an infected pin site will raise the skin temperatures that can be detected with thermography [6]. Regular monitoring of skin surface temperatures at pin sites might be a method for surveillance and contribute to early diagnosis of infections [4], [5]. At this exploratory stage of investigating pin sites with thermography, Orthopedic researchers are looking at the reproducibility and reliability of thermography as a surveillance tool [2].

Locating and tagging the Region of Interest (ROI) in an image is the first initial step for training a supervised learning model or extracting desired information in ROI. Annotation tools like VOTT (Visual Object Tagging Tool) [12], developed by Microsoft, allows manual tagging on Visible images. [17] used Amazons Mechanical Turk crowd-sourcing tool for annotating their visible images to later train their CNN model. As existing annotation tools are mainly designed aiming the supervised machine learning models, many application specific annotations tools [18], [19] are evolving in recent times.

Visible images usually have higher resolutions compared with the thermal and the radiometric images. Annotations on visible images using VOTT tool [12] or any other available tools like Amazons Mechanical Turk crowd-sourcing tool cannot be directly mapped onto thermal or radiometric images. Tools already designed to extract temperature data from thermal images has many limitations in terms of time, resources and data management. An example is the FLIR TOOLS [15] that provides the feature of extracting maximum, minimum and average temperatures at the ROI mapping visible and thermal images, but the user must manually draw a circle or square assuming the radius [2]. This inconsistency of manually assuming the radius makes the temperature information ex-

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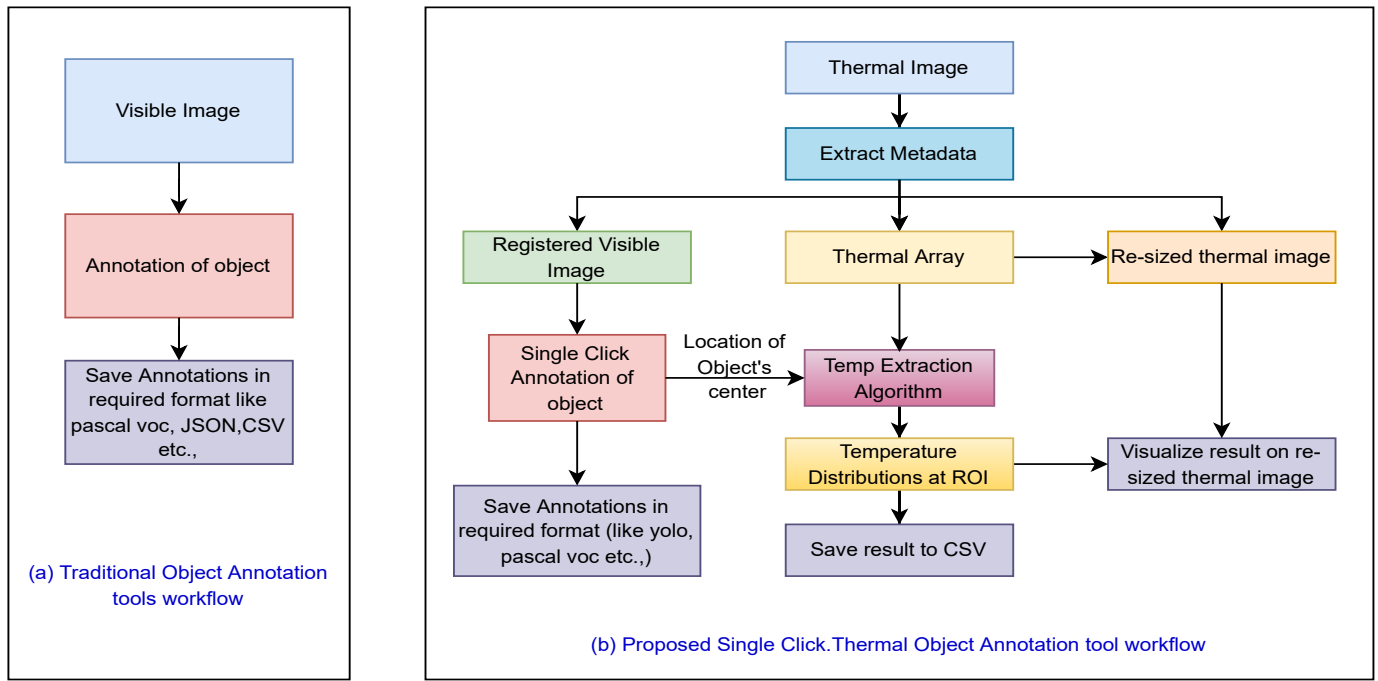


Fig. 1. Workflow of traditional and proposed annotation tools.

tracted from a large dataset less reliable especially for medical research. It is also not designed as an annotation tool and so does not support saving back annotations for future training of deep learning-based object detectors.

Additionally, keeping track of the temperature data at each pin site in each image over time is challenging since an external fixator has a unique construct for every patient with multi planar possibilities of several rings, pins and wires in each plane [8]. The complexity of this construct imposes a necessary step of labelling each pin with an unique code for easy identification that include information about the location of the pin in the external fixator and anatomical location. Automatic detection of the pin site of interest when multiple pins are visible in the image is challenging even with an object detection algorithm like YOLO [9].

Thermal pin site detection algorithm [1] was previously developed to detect pin sites on digital images and transfer ROI to their respective thermal images. This algorithm uses YOLOv5 [13] with modifications in the regression part and focuses to improve the precision of detecting the pin site. The recall is poor when exposed to new datasets obtained with different thermal cameras at this current stage and the algorithm aims to detect all the pin sites in an image, where only one of the pin site is of the interest for temperature extraction at this stage.

This current study aims to develop a tool that overcomes all these challenges to support orthopedic research in extracting temperatures at ROI, for further investigations on how those data are correlated to clinical finding and to train an evidence based deep learning algorithm. The overall objective is to understand how thermography can be used as a surveillance

tool for infection detection after surgery.

Figure 1 illustrates the workflow of a traditional annotation tools available in the market versus the workflow of the proposed single click thermal annotation tool for orthopedic pin site.

II. METHODS AND MATERIALS

The image dataset with pin sites are collected by the trained Research Nurse from Aalborg University Hospital with FLIR T540 camera. This is a cross sectional dataset and any information about the patient are not considered for developing this tool. The FLIR T540 camera captures both visible image and their respected thermal image. The images were with high quality and were acquired in a controlled environment in the hospital taking all the measures to avoid bias with absolute temperatures [1], [2]. Each study id has images of all pins from a single patient. Each image is captured aligning the pin site of interest to the center of the image and by calibrating manually in FLIR camera. Calibration metrics are needed for registering thermal part of visible image. The proposed tool relies on exiftool [16] for extracting meta information from thermal images. Exiftool [16] reads meta data of many digital and thermal cameras available in the market. The meta data of FLIR Thermal image can be obtained with exiftool [16] directly or with flirimageextractor tool [14]. This metadata provides all the necessary information about these calibration metrics.

The tool takes thermal image as input. This input is processed with flirimageextraction tool [14] to extract metadata. The metadata of the thermal image consists of all the necessary information about the calibration parameters for obtaining

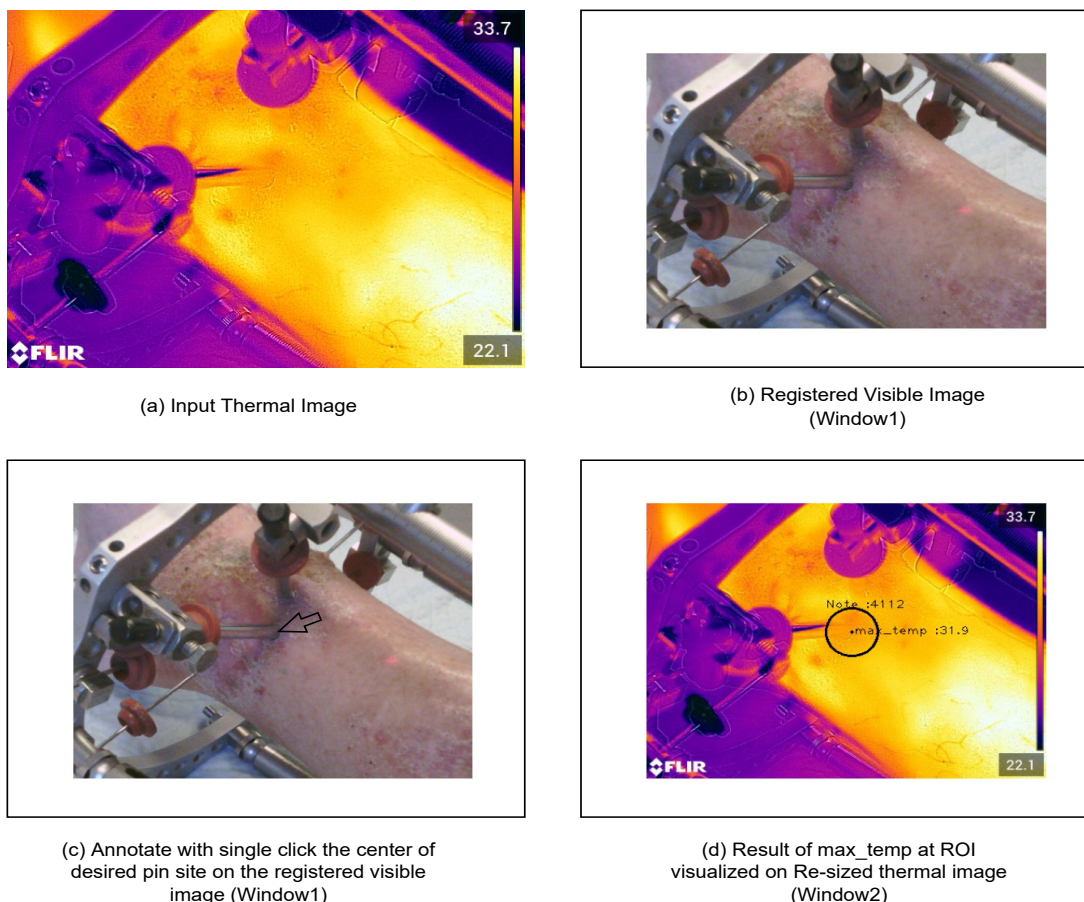


Fig. 2. Illustration of the tool with an example

study_id_121		
file_name	UPIN	max_temperature in ROI
FLIR4328.jpg	1112	25.75
FLIR4330.jpg	2112	26.81
FLIR4332.jpg	3112	26.79
FLIR4334.jpg	4112	26.94
FLIR4336.jpg	5212	29.61
FLIR4338.jpg	6212	29.86
FLIR4340.jpg	7312	29.72
FLIR4342.jpg	8312	30.63

Fig. 3. Table with maximum temperature information at ROI of all pin sites in the study id 121 extracted and saved as a csv file

registered visible image. This registered visible image is used for dynamic annotations. The algorithm asks clinicians for specifying radius. The default radius is 30 pixels. The ROI is the circle with specified radius from the pin site. Window1 with registered visible image pops up asking the clinicians to

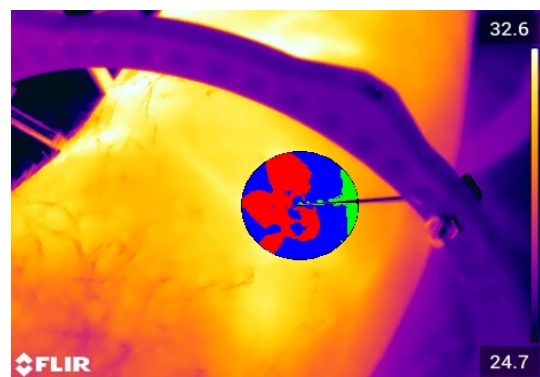


Fig. 4. Three level contour representation of ROI with radius 50 pixels from the pin site

mark the pin site of interest. The next step is the generation of temperature information in the required format. Window2 is the re-sized thermal image with the ROI and the pin site marked with the result. Figure 2 explains the steps of the proposed annotation tool with an example. The next sub section will elaborate further how this tool could be helpful to the research towards developing post surveillance systems

study_id	file_name	temp_center	max_temp_Rt	room_temp	sd	avg	pin_no	pin_or_wire	frame	loc	param1	param2	param3	param4	param5	param6	param7	...	param2808	param2809
127	FLIR4540.jpg	28.7	29.9	23.0	0.8	28.5	1	1	1	1	29.0	29.0	28.9	28.9	28.9	28.8	28.8	...	28.7	28.7
127	FLIR4542.jpg	30.6	31.7	23.0	0.8	30.7	2	2	1	1	31.1	31.1	31.1	31.2	31.2	31.2	31.3	...	31.3	31.4
127	FLIR4544.jpg	29.5	32.6	23.0	0.9	31.7	3	2	1	1	31.7	31.7	31.7	31.7	31.7	31.7	31.6	...	31.6	31.6
127	FLIR4546.jpg	31.0	31.9	23.0	3.4	29.0	4	2	1	1	23.3	23.3	23.3	23.4	23.4	23.4	23.4	...	23.3	23.3
127	FLIR4550.jpg	31.4	31.9	23.0	2.1	30.4	5	2	1	1	23.5	23.6	23.7	23.6	23.6	23.6	23.6	...	23.6	23.6
127	FLIR4552.jpg	31.2	32.1	23.0	1.0	31.1	6	2	1	1	31.3	31.3	31.3	31.4	31.4	31.4	31.5	...	31.6	31.6
127	FLIR4554.jpg	28.0	32.6	23.0	1.1	31.6	7	2	1	1	32.0	32.0	32.0	32.1	32.1	32.1	32.1	...	32.1	32.1
127	FLIR4556.jpg	31.2	32.2	23.0	0.9	31.1	8	2	1	1	30.8	30.8	30.7	28.1	25.7	27.4	30.9	...	31.2	31.4
127	FLIR4558.jpg	30.5	31.5	23.0	1.1	30.1	9	2	1	1	30.2	30.2	30.3	30.4	30.4	30.5	30.6	...	30.7	30.7
127	FLIR4560.jpg	32.4	33.5	23.0	0.6	32.7	10	1	2	2	32.8	32.8	32.7	32.7	32.7	32.6	32.6	...	32.5	32.5
127	FLIR4562.jpg	32.9	34.5	23.0	1.1	33.2	11	1	2	2	33.4	33.3	33.3	33.3	33.3	33.3	33.2	...	32.9	33.1
127	FLIR4564.jpg	30.8	32.2	23.0	1.1	31.1	12	1	3	2	31.2	31.2	31.2	31.3	31.3	31.3	31.3	...	31.3	31.3
126	FLIR4476.jpg	33.0	34.6	24.0	0.8	33.0	1	2	1	2	34.6	34.6	34.6	34.5	34.5	34.5	34.4	...	34.3	34.3
126	FLIR4478.jpg	33.8	34.7	24.0	1.2	33.3	2	2	1	2	32.6	32.7	32.7	32.7	32.7	32.7	32.7	...	32.8	32.8
126	FLIR4480.jpg	33.5	34.9	24.0	1.5	33.2	3	2	1	2	32.5	32.6	32.6	32.7	32.7	32.7	32.8	...	32.6	32.7
126	FLIR4482.jpg	33.3	34.5	24.0	1.4	33.0	4	1	1	2	28.2	28.4	28.4	28.5	28.6	28.4	27.1	...	26.3	26.1
126	FLIR4484.jpg	32.8	33.5	24.0	1.1	32.5	5	2	1	2	32.6	32.6	32.5	32.5	32.5	32.5	32.5	...	32.5	32.5
126	FLIR4486.jpg	33.3	34.1	24.0	0.4	33.2	6	2	1	2	32.7	32.7	32.7	32.7	32.7	32.7	32.7	...	32.7	32.7
126	FLIR4488.jpg	33.0	35.1	24.0	0.6	34.5	7	2	1	2	34.8	34.8	34.7	34.7	34.7	34.6	34.6	...	34.5	34.5
126	FLIR4490.jpg	29.0	35.2	24.0	2.2	32.3	8	1	2	2	28.1	28.1	28.1	27.9	27.9	27.9	27.9	...	27.8	27.8
126	FLIR4492.jpg	33.4	34.9	24.0	1.5	33.5	9	1	3	2	27.8	29.9	31.8	32.6	33.2	33.5	33.9	...	34.1	34.1

Fig. 5. Temperature information of 2809 pixels in the ROI with 30 pixels radius from all pin sites in study id 126 and 127.

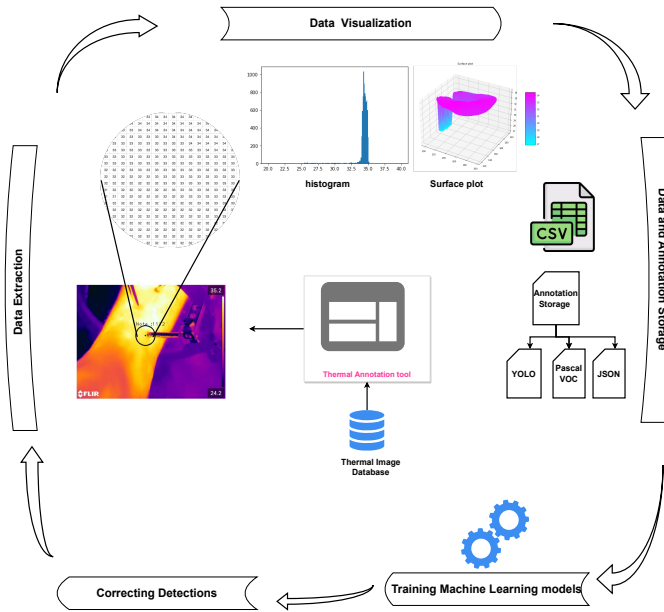


Fig. 6. Figure explaining potential usage of thermal pin site annotation tool

for infection.

A. Extraction of Maximum temperature in the ROI

This tool is mainly designed to support the study [2] to observe the relation of maximum temperature at ROI with infection grade. This tool takes pin site annotation dynamically from window1. The tool then draw a circle around the pin site with the specified radius and extracts location of pixels in the image. The temperature values in these locations are mapped to respective radiometric image. Maximum temperature in the ROI is identified and displayed on the re-sized thermal image for the clinician with Unique Pin Identification Number (UPIN). The tool further saves all these information in a csv sheet for future analysis.

B. Temperature Contour extraction in the ROI

Numerous studies used temperature distributions to estimate size, shape and location of abnormal tissues [10], [11]. Temperature contour extraction assists researchers to visualize ROI with color gradients based on temperature. These contours assist researchers to observe the thermal gradients within the ROI. These temperature contour visualization for longitudinal data facilitate clinicians and researchers to acquaint with the shape and direction of increased thermal activity around the pin site. Figure 4 illustrates three level contour with a temperature difference of 0.5 at each level. The pixels in red represents maximum threshold temperature in the ROI and pixels in green corresponds to minimum threshold temperature in the ROI. The pixels in blue are with temperature values between maximum and minimum thresholds.

C. Temperature distribution extraction in the ROI

Data acquisition is the crucial step for training machine learning models. Annotating and Extracting temperature distributions in a csv format can further assist researchers in their research towards training and developing automated tools for early diagnosis of infections through machine learning. The temperature distribution of ROI with 30 pixels radius from pin site has 2809 pixels temperature information. Figure 5 is a csv with temperature information of 2809 pixels in the ROI with 30 pixels radius from pin site.

III. DISCUSSION

Manually annotating thermal images may be challenging and can lead to human errors which might not be acceptable in medical applications. In many thermal images, the exact pin site location is not apparent due to radiation. This difficulty could be clearly noticed in Figure 2(a). To avoid the bias caused due to thermal image annotations, the tool provides the flexibility for clinicians to annotate registered visible image. The flexibility to annotate on registered visible images can assist to reliably map these annotations to their respective re-sized thermal or radiometric images.

The proposed tool overcomes the challenges of matching the pin site of interest in multiple detection's by deep learning-based object detection algorithms. While capturing images for the study, extra measures are taken to align the pin site of interest to the center of the image. There might be a possibility to have multiple pins in an image but the center pin site is always the pin site of interest. Example in Figure 2 has two pin sites in the image out of which only pin site at the center is of interest. Each pin site of interest is given a Unique Pin Identification Number (UPIN). In figure 2, the pin site with note (UPIN) - 4112 is the pin site of interest. The number of fields in the UPIN differs with the pin number. The pin number can be a single or a double digit. This UPIN provides us details of the location of pin site. So detection of pin site that matches with this UPIN is of high importance to avoid bias with technical errors in diagnosis. Consequently, developing a tool which allows clinicians to annotate pin site of interest reduces the risk of poor recall rate with object detectors.

As object detection is a supervised learning problem, annotating data for training is time consuming. These annotations should be done by a trained professional for avoiding bias. So there should be flexibility for the tool to re use these annotations for training pin site detectors while collecting research information.

Despite the fact that the data is cross sectional at the moment, the tool may in the future advantageous to monitor maximum temperature and temperature contours in the ROI once the longitudinal data is available.

This tool allows dynamic annotation of the pin site of interest to retrieve the thermal information with in the specified radius. Evolution of user friendly annotation tools for automatic temperature extraction could assist researchers look deeper into the usage and reliability of thermography in early diagnosis of infections with External Fixation. This tool can be further integrated with the previously developed Thermal Pin site Detection Algorithm [1] for retrieving the information from missing detection's as well as re-training the model for performance optimizations with new datasets.

Figure 6 explains the potential usage of the proposed annotation tool towards developing automated tools for early diagnosis of infections.

This tool utilize calibration matrices extracted from meta-data of thermal image. Efficient mapping of region of interests depends on accuracy of metadata. The region of interest should be in the center for efficient mapping of regions.

CONCLUSION

In this paper, we presented a user friendly research tool to annotate orthopedic pin site of interest to retrieve temperature distribution in the Region of Interest (ROI). To our knowledge, this is the first annotation tool dedicated to thermal images with wounds. Our work for developing the single click annotation paves a way for future research in developing automated wound care AI tools for home based surveillance of infections through thermography.

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